

Japanese physicist Hideki Yukawa's Nobel Prize-winning work advanced our knowledge of two of the four basic forces in nature—the strong and weak nuclear forces. He proposed the existence of a new kind of subatomic particle, the meson, to explain the actions of protons and neutrons inside atoms.

In 1935, while a physics lecturer at Osaka Imperial University, Hideki Yukawa published a pioneering paper on nuclear forces. The proton had been discovered in 1919, and the neutron in 1932, and physicists knew that atomic nuclei contain both particles. However, it was not clear how the positively charged protons were able to interact within the nucleus without repelling one another. Yukawa's theory predicted that the protons and neutrons interacted by exchanging an intermediary "carrier" particle, which was later called a meson.

#### Meson's existence confirmed

In 1937, a team of US scientists discovered a new particle in cosmic radiation which had a mass close to that of Yukawa's predicted meson. This appeared to confirm Yukawa's theory, and his reputation spread beyond Japan. However, it was later discovered that a different particle, now called the pion (or pi-meson), was involved in binding the nucleus instead. With Yukawa's work on nuclear forces confirmed, he was awarded the Nobel Prize in Physics in 1949.

#### MILESTONES

##### PREDICTS PARTICLE

Claims in his 1935 paper that mesons carry special properties that hold the atomic nucleus together.

##### AWARDED NOBEL PRIZE

Becomes Japan's first Nobel Laureate in 1949 when awarded the Physics Prize for his meson discovery.

##### BECOMES DIRECTOR

Appointed first chairman of Japan's Research Institute of Fundamental Physics in 1953.

**"Nature is simple  
in its essence."**

Hideki Yukawa, 1959

*In 1949, Yukawa became professor of physics at Columbia University in New York, where he taught a graduate course in meson theory. The blackboard he used there was moved to Osaka University in 2014.*